files will tell you the time taken for the build.

You can interrupt the build process in the middle if you're running out of disk space. Free up some disk space and then run make again to resume the build process. If some missing packages are reported in the middle (which happens very rarely), install them and run make again to resume.

Requirements

- A 1.5 GHz or higher processor
- 512 MB RAM; 1 GB or higher recommended to minimise build time
- 10 GB free disk space (just for build, excluding OS requirements), 5 GB or more space for further hacks and if you plan to add debug support for some modules.
- A distro with a comfortable package manager
- Internet connectivity on Linux recommended

Step-by-step procedure

1. **Check out ooo-build:** You can check out the *ooo-build* repository through version controls like SVN, Git, etc:
   ```
   svn co svn://svn.gnome.org/svn/ooo-build/trunk ooo-build
   ```
   or:
   ```
   git clone git://anongit.freedesktop.org/git/ooo-build/ooo-build
   ```
   You can also manually download the latest *ooo-build* tarball from *download.go-oo.org*. However, using a version control is always advisable.
2. **Configure:**
   ```
   ./autogen.sh  --with-distro=YOURDISTRO --with-gcc-speedup=ccache,icecream
   ```
   Here you have to specify your distribution name with the *--with-distro* option. To see a list of supported distros, run the script once without this option. Use the nearest match if a specific distro is not listed.

   The *ccache,icecream* option helps to accelerate the build process. The build time will substantially reduce if you add those with the *--with-gcc-speedup* option. Use *--with-icecream-bindir=/usr/bin* if you get an error like "icecream's gcc not found". If you haven't used version control and downloaded the tarball directly, use the configure script instead of *autogen.sh.*
   ```
   ./configure  --with-distro=YOURDISTRO --with-gcc-speedup=ccache,icecream
   ```
3. **./download:** This will download the required sources directly using *wget*. Please note that the resume option is not available, as the download script won't use *wget* with resume option. If your Internet connection is slow or not available you have to download the required sources manually using a different machine and copy to *src/* directory. Apply your own logic to find out the required sources. Use the *--with-tag* option during configuration to use different sources other than the default.
4. **make:** This will start the build process, starting with extracting source files, applying patches, and then going on to the actual build (configuring and compiling each module). Wait until the actual build starts, as many missing packages will be reported at this phase. For example, if an error like '*cups.h* not found' appears, install the *cups-devel* package to fix it. Once a few modules (*boost, stlport*, etc) start compiling, you can take a break until the build completes.
5. **bin/ooinstall -l <path-to-install>:** It installs OOo with links from the built sources—that's why it's called a 'linkoo' build. You can also use the *make install* command to install your OOo build. However, after every hack you have to run *make install* again in order to reflect the changes. So the special script *ooinstall* can be used instead to reflect changes without a reinstall every time.
6. **cd <path-to-install>/program**
7. **source ./ooenv:** This will prepare a shell to run OOo.
8. **./soffice.bin:** You can also add options like *-writer, -calc*, or *-impress*, to run specific components directly. The *soffice* command takes the file name as the argument, which is loaded on start-up.

Hacking Go-oo

Now that the *soffice* instance is running properly from the built source, let's do a small hack. Go to the *vcl/* directory [*ooo-build/build/ooo3??-m??/vcl/*] and edit *source/window/menu.cxx* by changing the following line (around line number 1800):

```
pData->aText = rStr;
```

to:

```
pData->aText = String(rStr).Reverse();
```

Run the following commands to build the *vcl* module:

```
. ../LinuxX86Env.Set.sh ## dot space dot dot slash LinuxX86Env.Set.sh
build
```

You can use *build debug=true* instead to add debugging support for a specific module.

We'll use the *deliver* command on completion of the build to ensure changes are made when, by chance, *make install* is run at a later time.

Now, make sure no previous instance is running by closing OOo manually or with the command:

```
killall -9 soffice.bin
```

Re-run *soffice.bin* from the program directory to see the changes. Notice anything different? Yes, all the menu item characters are now in the reverse order (Figure 2).